CFAI MIDI-PYRENEES
INSTITUT CATHOLIQUE DES ARTS & METIERS
Site de Toulouse

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Anode-Covering Technology Applied to Impressed Current Cathodic Protection

Valentin THIERS
Year 2018

Icam supervisor : ***Mrs. Corinne Curvale***

Apprenticeship master : ***Mr. Stéphane RAMON***

Dalhousie University supervisors :

Dr. George Jarjoura
Mr. Farzad Hashemi

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Abstract

In order to better understand the behaviour of the impressed current cathodic protection using a covering anode, samples of metal to be protected have been prepared and exposed to a corrosive middle. Several mixtures of covering anode were tested in order to obtain the best compromise between the mechanical and electrical properties. The resistivity of these coatings were measured thanks to several methods. At a later point, an experimental method was designed in order to measure the efficiency of the cathodic protection and determine the best electrical parameters. The advantages of this protection method were highlighted and its application in the industry evaluated.

Résumé

Dans le but de mieux comprendre le comportement des systèmes de protection cathodique par courant imposé utilisant un revêtement anodique, des échantillons de métal à protéger ont été préparés et exposés à un milieu corrosif. Différentes mixtures de revêtement anodique ont été testés afin d'obtenir le meilleur compromis entre les propriétés mécaniques et électriques. La résistivité de ces revêtements a été mesurée à l'aide de différentes méthodes. Par la suite, une méthode expérimentale a été mise au point afin de mesurer l'efficacité de la protection cathodique et de déterminer les meilleurs paramètres électriques. Les avantages de cette méthode de protection ont été soulignés et son application dans l'industrie évaluée.

1. Introduction

1.1. Background

Aqueous corrosion of metals is a problem which has been faced since antiquity.[1] When it occurs on a metal surface, the material loses its mechanical properties and the consequences can be serious. In the oil and gas industry, the pipelines are made of carbon steel, and they are buried in soil or immersed in water. If some corrosion areas appear on such a structure, it may crack and generate environmental and financial issues. The most efficient way that human beings have found to protect a metallic structure from corrosion is to isolate it from any corrosive environment while coating it with a corrosion resistant material. The first coatings were made with bitumen; then, this technology evolved and the structures were coated with paint. Nowadays, thanks to electroplating and electroless coating technologies, the structures are covered with a thin corrosion resistant metal layer.[2]

However, many exterior elements can deteriorate the coating with time. Consequently, another type of protection is necessary to protect the non-coated areas. One of the most common methods used in the twenty-first century is the Impressed Current Cathodic Protection (ICCP). It offers satisfying results in terms of protection, therefore the installation and the working costs are high: anode beds have to be buried in the ground along long distances and a huge surface of soil needs to be surveyed.[3] The design of such protection systems must take into account a large panel of parameters such as the soil or water resistivity, the shape of the structure to be protected and the potential surrounding structures. It also needs four mains elements: a structure to be protected called cathode, a remote counter-electrode called anode, a conductive middle as soil or water called electrolyte, and an electrical circuit fitted with a suitable rectifier. This results in a large amount of limiting parameters that have to be taken into account for the design of an ICCP installation. Also, the electrolyte generates the two unsolved issues of this method: the stray currents present in the soil or water[4] and the hydrogen generated damages.[5]

Protect a metallic structure thanks to the ICCP method while being independent of the electrolyte is an important innovation axis in the field of corrosion protection. It would fix those two problems and allow to protect the atmosphere exposed structures such as cars chassis. The solution proposed by Mr. Hashemi in his patent is the utilisation of a conductive layer of coating as covering anode. It makes the ICCP method applicable no matter what the surrounding middle is, and it has financial and technical advantages compared to the common method. The solution to free the system of the electrolyte was compared to

[1] Baeckmann, W. von, Schwenk, W., Prinz, W., Houston (1997); p. 2
[2] Ibid; p. 4
[3] Parker, Marshall E., Peattie, Edward G., Houston (1984); p. 66
[4] Ibid; p. 100-108
[5] Roland Tolulope Loto, Roy Olakunle Loto, Cleophas Akintoye Loto (2016); p. 372

other solutions using different technologies; it was demonstrated that using a covering anode in ICCP is the only way to protect a structure efficiently against corrosion in an atmospheric environment.[6]

1.2. Objectives

The objective of this research is to design an experimental setup and conduct tests to prove this new technology in particular conditions. It was divided into three main steps:

- First, there was a need to create a product suitable for this experiment: a metal piece to be protected fitted with a layer of insulating primer and a covering anode. Also, a good composition of materials to create a suitable covering anode and applying it had to be found.
- Second, tests were conducted to know the resistivity of the material and to determine the electrical parameters that the rectifier must supply to give a good level of protection to this specific system.
- Lastly, a procedure was written to allow future researchers to recreate the same kind of covering anode and lead the same kind of tests.

This report can then be considered a starting point for futures researches about the ICCP using the covering anode technology.

1.3. Thesis outline

This paper is divided into four chapters. Chapter 1 introduces the field of research. Chapter 2 is a comprehensive literature review on aqueous corrosion of metals and on ICCP. The working principles of the ICCP common method and the covering anode method are also explained in this chapter. The covering anode materials, the application process and the experimental setup are discussed in Chapter 3. Chapter 4 is allocated to present the obtained results. Lastly, the experience and results are discussed in a conclusion and future work is proposed.

2. Literature review

2.1. Aqueous corrosion of metals

Corrosion can occur under many conditions and take several forms. However, when it occurs in an aqueous environment such as salt water, soil or atmospheric moisture, it is generally due to two or more electrochemical reactions.[7] These reactions drive the atoms from their metallic form to their ionic forms

[6] Jarjoura (2017)
[7] D.W. Shoesmith (1992); p. 45

which are more stable; they result in loss of volume in the metal and a loss of its mechanical properties. In aqueous corrosion of metals, at least one of these reactions is an oxidation reaction (Eq 1) and at least one is a reduction reaction (Eq 2). They can be written as follows:

Oxidation reaction:	$M_{(s)} \rightarrow M_{(aq)}^{n+} + ne^-$	(Eq 1)
Reduction reaction:	$Ox + ne^- \rightarrow Red$	(Eq 2)
The overall reaction:	$M_{(s)} + Ox \rightarrow M_{(aq)}^{n+} + Red$	(Eq 3)

In the common cases of buried or immersed carbon steel structures, more than one reduction reaction occurs. The most important reactions in common cases are as follows:

Oxidation reaction:	$Fe_{(s)} \rightarrow Fe_{(aq)}^{2+} + 2e^-$	(Eq 4)
Reduction reaction:	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	(Eq 5)
	$2H^+ + 2e^- \rightarrow H_2$	(Eq 6)
Overall reactions:	$2Fe_{(s)} + O_2 + 4H^+ \rightarrow 2Fe_{(aq)}^{2+} + 2H_2O$	(Eq 7)
	$Fe_{(s)} + 2H^+ \rightarrow Fe_{(aq)}^{2+} + H_2$	(Eq 8)

Each oxidant-reducer couple has a free potential measured with the hydrogen reduction reaction's potential as reference (Eq 6). This free potential determines the corrosion resistance of a metal in an aqueous environment. Indeed, the more negative the free potential of the element, the more favoured its oxidation reaction (Eq 4), which is corrosion.

The electron flow generated by the oxidation reaction corresponds to the anodic current. In the same way, cathodic current corresponds to the electron flow generated by the reduction reaction. In a free state of corrosion, those currents are equal and depend on the potentials of each couple of oxidant-reducer. We have: $|i_a| = i_c = i_{corr}$. Where i_a is the anodic current density, i_c is the cathodic current density and i_{corr} is the corrosion current density. The corrosion rate is the rate at which the metal dissolution reaction takes place; it is directly related to the anodic current. Decreasing the value of the anodic current on a surface is decreasing the corrosion rate of this surface.

2.2. Impressed Current Cathodic Protection : Theory

The working principle of cathodic protection can be explained by the mixed potential theory. It intends to predict the anodic and cathodic currents densities in a system in function of the potentials of the couple

oxidant-reducer of the different elements present. *Figure 1* shows the currents which appear on the metal surface when a piece of solid iron is immersed into a hydrochloric acid (HCl) solution.

The equilibrium potential of the system E_{corr} is above the free potential of the couple Fe/Fe^{2+} versus a Standard Hydrogen Electrode (SHE): $E_{Fe/Fe^{2+}}$. That makes the piece of iron become anodic to its environment. The metal will then act as an anode which means that it will corrode (Eq 4). Furthermore, *Figure 1* shows that the corrosion rate is more important for the iron when it is immersed in HCl solution than its corrosion rate in presence of its own ions. Indeed: $i_{corr} = i_a > i_{0\,Fe/Fe^{2+}}$.

The ICCP is a technique used to prevent corrosion which can be understood thanks to the mixed potential theory. The working principle of ICCP consists in reducing the corrosion rate of the metal to be protected by shifting the potential of a system to a more negative potential. It is called cathodic polarization of the metal. *Figure 2* shows how the cathodic polarization of the metal significantly reduces the anodic current on the metal surface, note that the current axis is plotted in a logarithmic scale.

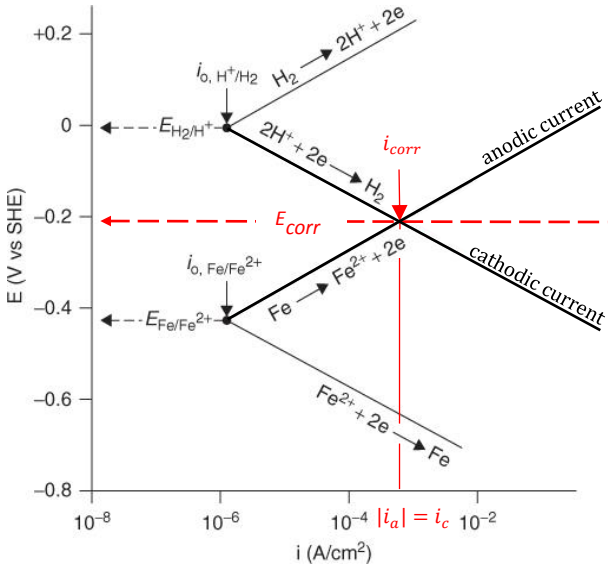


Figure 1 : Evans diagram of iron into an acid solution

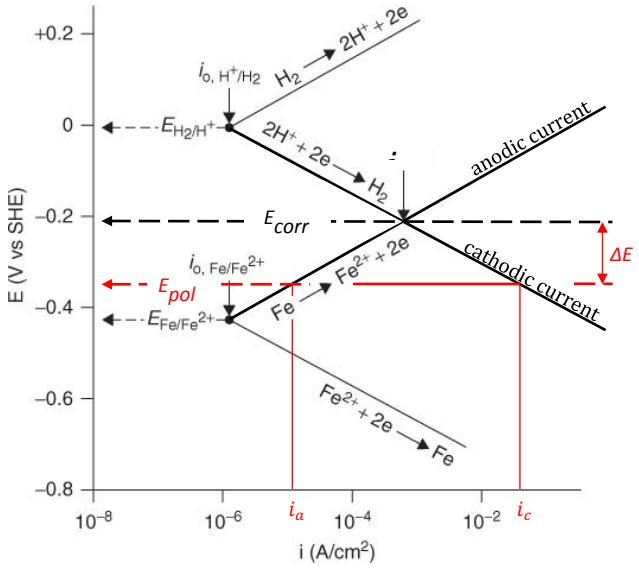


Figure 2 : Evans diagram of polarized iron electrode in an acid solution

This polarization can be done by sending a current into the electrolyte to the structure to be protected thanks to an anode. This increases the cathodic current flowing into the electrolyte and reduces the anodic current on the iron surface. However, the anodic current on the anode surface is significant which is why the materials used as anodes in ICCP are naturally resistant to corrosion. These can be graphite, lead, conductive polymer or even ceramic.[8]

As the current axis is plotted in a logarithmic scale, shifting the potential to a total protection potential (where: $i_a = i_{0\,Fe/Fe^{2+}}$) would require providing a large amount of current, and it would generate high protection costs. The estimated potential to protect a structure in soil was chosen to be -850mV versus a

[8] Robert H. Heidersbach (1992); p. 469

copper-copper(II) sulfate electrode (Cu/Cu^{2+}).[9] However, if the potential is more negative than -1.2V, hydrogen generated damages can occur. They are due to the presence of hydrogen bubbles on the cathode surface; when those hydrogen bubbles accumulate under the coating it creates high pressure areas which result in a blister on the coating. Embrittlement of the metal is another form of hydrogen generated damages.

2.3. Activation polarization and concentration polarization

When the potential of a system is shifted to a lower value, several phenomenon are taking place before being in a stable state of polarization. Those phenomenon are called activation polarization and concentration polarization. They have a high impact on the current flowing through the system, therefore, on the corrosion rate.

- Activation polarization defines the energetic barrier for a positive charge to exchange current with the metal surface. It occurs as soon as the structure becomes cathodic. This barrier is due to the middle transporting the electrical charges, which is often water molecules. Indeed, in the electrolyte, a positive charge is surrounded by the H_2O molecules which are dipoles. The water molecules carry the positive charge up to the cathode surface. Once at the cathode surface, the driving force (or potential) must be large enough to extract the positive charges form the water molecules and make them exchange current with the metal.
- Concentration polarization defines the energetic barrier encountered into the electrolyte. Indeed, the rate at which the metal consume the positive charges is higher than the rate at which the positive charges can flow from the electrolyte to the metal surface. Therefore, the concentration of positive charge directly around the metal surface, few minutes after the activation of the protection, will be lower than the concentration of positive charges in the surrounding electrolyte.

These barriers are represented in *Figure 3* below.

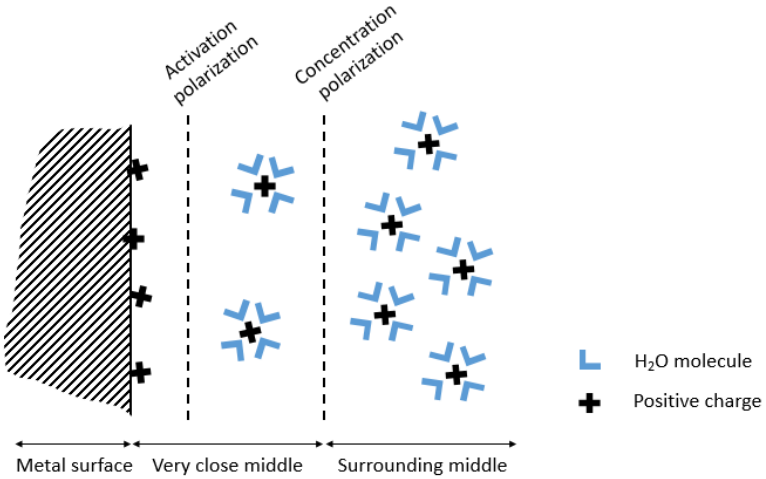


Figure 3 : Representation of activation and concentration polarization

[9] Ibid; p. 467

2.4. Impressed Current Cathodic Protection : Practice

This protection method is widely used in many industries, however, its application requires huge construction means and studies. In order to catholically polarize the structure to be protected, a set of remote anodes has to be buried in the ground, in the case of onshore structures, which generate high installation costs. Moreover, to face the voltage drop around the anode, it needs to be installed in a low resistivity material which helps to spread the current all around the anode. This is called backfill, and it can be made of bentonite clay or other conductive porous materials.[10] *Figure 4* is a schematic representation of an ICCP installation: the pipeline to be protected ① is electrically linked to the negative terminal of the DC source ④, and the remote anodes made of corrosion-resistant materials ② surrounded by their backfill ③ are electrically linked to the positive terminal of the DC source. The structure to be protected and the anode set are buried in the electrolyte ⑤.

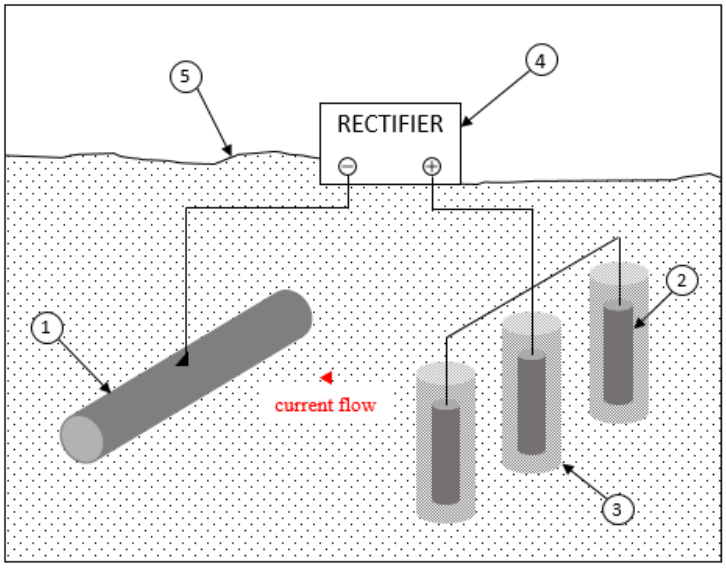


Figure 4 : Schema of an ICCP installation to protect a burried pipeline

The rectifier provides a current which flows into the electrolyte and cathodically polarize the pipeline while making it less anodic than the buried anodes. Thanks to this, the reduction reaction will mainly take place on the pipeline surface and the oxidation reaction will mostly take place on the anode surface, which is a noble metal with high resistivity to corrosion[11].

This kind of installation requires huge installation means. Indeed, anodes sets must be buried all along the structure to correctly protect the entire surface. In the case of a pipeline, the voltage drop along the electrolyte will lead to an underprotected area if all the anodes are buried in the same place. *Figure 5* below shows how the engineers have to design the location of the anodes to keep all the pipeline in a good protection state.

[10] Ibid; p. 466
[11] Tan Yongjun, (1963);

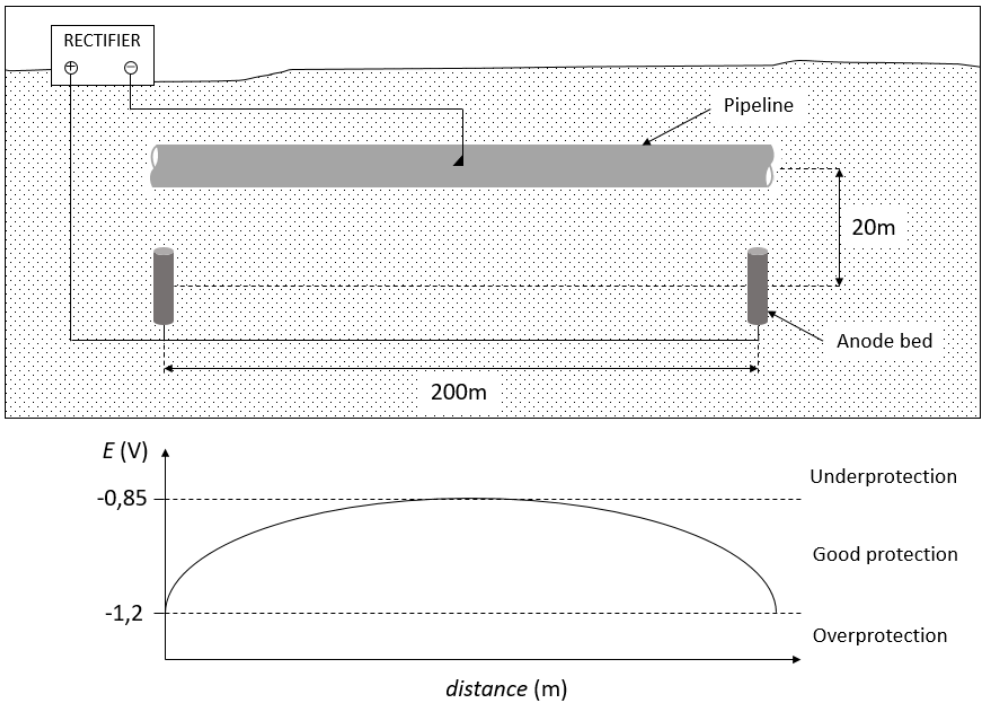


Figure 5 : Polarization along a pipeline (distances are arbitrary here)

Another problem encountered by those installations are the presence of stray currents in the electrolyte. They are exterior currents flowing through the electrolyte in random directions. It may cause the corrosion of other structures located around the installation or even cause corrosion of the structure to be protected. Those stray currents can be generated by industrial installations as welding workshops, electric railway as tramway, or by natural phenomenon as lightning. They are unpredictable and inevitable.

2.5. Utilization of a covering anode for ICCP

The innovation consist in having an anode all around the structure to be protected instead of having sets of remote anodes. It can be done by covering the coating with a conductive material such as a carbon filled polymer. The anode and the structure are linked to a rectifier in the same way used in the common method, so while the coating is intact, there is no electrical contact between them. However, a crack in the coating will allow a thin part of the electrolyte (soil, water or moisture) to electrically connect the anode to the cathode and switch on the protection. The distance between both electrodes is the thickness of the coating which is usually less than 1 mm.

The utilization of a covering anode in ICCP has many advantages. First, it naturally solves the problem of the stray currents by acting as a Faraday cage.[12] Moreover the electrodes are relatively close which allows moisture or splash to electrically connect them and avoids voltage drops into the electrolyte. Therefore, it significantly reduces the costs of installation (the anode beds are not dug anymore) and

[12] Faraday cage: Enclosure made of conductive material which prevent electrical currents to flow into it.

running (the electrical power consumed is lower). The working principle of the ICCP does not change, but the way it behaves is quite different than the common one. *Figure 6* is a schema of a part of a structure protected by the ICCP using the covering anode technology.

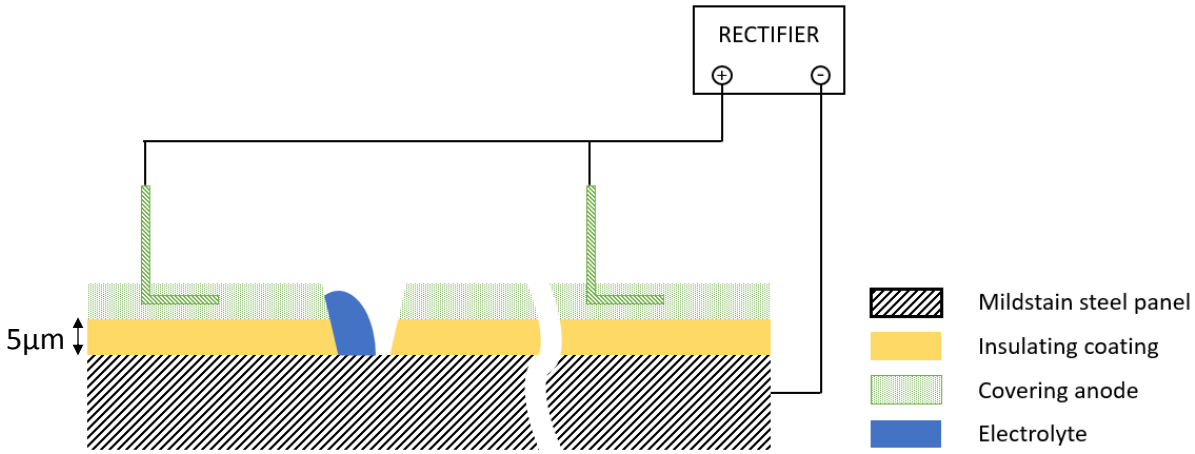


Figure 6 : Working principle of covering anode used in ICCP

The challenge to face for applying this type of protection on a structure is to find the best material to create the covering anode. The insulating coating can be made thanks to polyurethane, epoxy or paint and these products are available on the market at acceptable prices. However, cheap, corrosion resistant and good conductive coating materials are in the development phase. The low needs in currents are advantageous and a simple low power DC source can be used, it can be a car battery.[13]

Studies to find the advantages and disadvantages of the wide variety of possible couple matrix / filler can be conducted. Materials such as polymers, resins or adhesive can be used as matrices. The fillers can be powder or granulate of electrically conductive materials like carbon, graphite, zinc, nickel, and more.

3. Contribution

The experience represented in *Figure 7* is designed to study the behaviour of the ICCP while applied to the covering anode technology. A plate protected by the ICCP using a covering anode method was immersed in a corrosive middle for cycles of 24 hours. During every cycle, the potential of the plate was set versus a Saturated Calomel Electrode (SCE) thanks to a potentiostat, and its value was verified with a multimeter. The net current flowing into the circuit and the potential applied by the rectifier were recorded after 2, 10 and 24 hours.

[13] Hashemi, (2016); p. 6

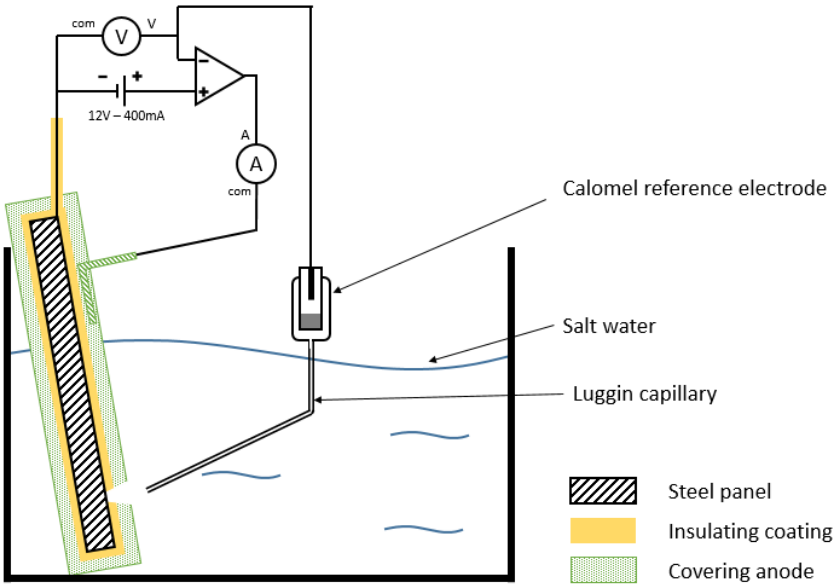


Figure 7 : Schema of the expreimental setup for an ICCP test

In this configuration, the potential between the SCE and the cathode remains at a fixed value thanks to the potentiostat which control the potential applied between the cathode and the covering anode. The imposed value for the potential between the SCE and the cathode can then be defined. A first test was done with an imposed value of -850mV versus the SCE, which is supposed to be the protection potential versus a SCE. Indeed the protection potential in salt water is given by the literature as -805mV versus a silver chloride electrode (Ag/AgCl).[14] The potential line below, *Figure 8*, shows the protection potential calculated versus a SCE.

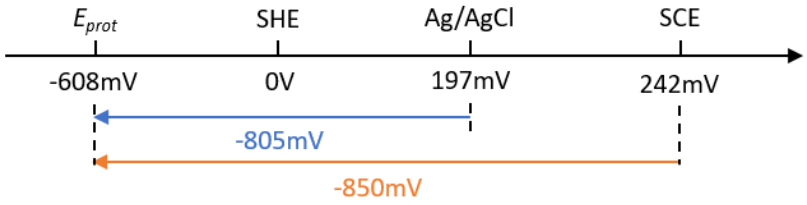


Figure 8 : Protection potential, E_{prot}, versus several references electrodes

To confirm the viability of the calculated protection potential, -850mV versus a SCE other tests with more and less negative potentials were run. These tests permit to frame the values of the protection potential between the values of the under and overprotection potentials.

[14] Robert H. Heidersbach (1992); p. 468

3.1. Preparation of the plate

- ### Soldering, cleaning and roughening.

The plate to be protected is made of AISI 1018 mild steel, and has the following dimensions 150x110mm and 1mm thickness. The rust is removed with a sandpaper of grid 120. Then an electrical wire is soldered to one of the corners of the plate thanks to a soldering gun and a metal work acid core solder. It is advised during this operation, to leave the soldering gun in contact with the plate for 10 seconds once the solder has melt, then to slowly remove the gun without generating any displacement between the wire and the plate. The plate is then degreased in an alkaline solution for 5 minutes at 80°C. The alkaline solution is composed of 2L of water, 38g of NaOH and 62g of dishwasher detergent. The final roughening is done while polishing the plate with the sandpaper of grid 380. The plate is shown in *Figure 9*. A wiremesh electrode must be prepared while welding a wiremesh square of 2x2cm to an electrical wire as shown in the *Figure 10*.



Figure 9 : Bare metal plate



Figure 10 : Wiremesh electrode

- ### Insulating coating

The first coating was made with the Interlux Primecote resin and hardener. Each side of the plate is coated with 4g of mixture (3g of resin for 1g of hardener) thinned with 1mL of acetone. The plate is placed in a suction hood for 8 hours for the coating to completely dry. Then, the face opposed to the soldering is roughened with a sandpaper grid 120 and a tape is applied on the edges of the plate to fully insulate it. This is shown in *Figure 11*.



Figure 11 : Primer coated plate

- ### Covering anode preparation and application

Several mixtures were tested as covering anodes to determine the composition which gives the best compromise between adhesion, consistency and conductivity. The quantities of the several components for 3g of covering anode are given in *Table 1* below and on the next page:

Sample number	**West System 105 epoxy resin** g - (ml)	**West System 206 slow hardener** g - (ml)	**Carbon powder** g	**Graphite powder** g	**Percentage of filler in weight** % wt.
0	1.79 - (1.54)	0.31 - (0.31)	0.45	0.45	30
1	2.20 - (1.90)	0.38 - (0.38)	0.15	0.30	15
2	2.10 - (1,81)	0.36 - (0.36)	0.18	0.36	18

3	2.04 - (1.76)	0.35 - (0.35)	0.20	0.40	20
4	2.02 - (1,74)	0.35 - (0,35)	0.21	0.42	21
5	1.94 - (1,67)	0.33 - (0.33)	0.24	0.48	24
6	1.91 - (1.65)	0.33 - (0.33)	0.25	0.50	25
7	1.87 - (1.61)	0.32 - (0.32)	0.26	0.54	27
8	1.79 - (1.54)	0.31 - (0.31)	0.30	0.60	30

Table 1 : Different covering anode compositions

The mixture is made by weighing the desired amount of powder, then a variable amount of acetone is added to solve the powder. It is necessary to have a kind of solid paste before adding the resin. Once the resin is added the mixture is stirred for two minutes and finally, the hardener can be added and mixed with the others products. After obtaining a homogeneous mixture, the covering anode is applied on the side of the plate which is opposed to the soldering. Two minutes after the application of the covering anode, the wiremesh electrode is gently pushed 1cm below the top of the plate and covered with the mixture, as shown in *Figure 12*. The plate is left for 15 hours in the suction hood to complete the first phase of curing, then it is placed during 8 hours in an oven at 55°C for the covering anode to finish to cure. Finally, a strike of 8 cm is made on the bottom of the plate, as shown in *Figure 13*, in order to obtain a surface of bare metal of around 1cm².



Figure 12 : Anode covered plate

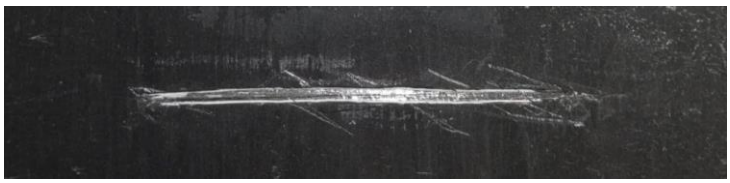


Figure 13 : Strike on the bottom of the plate

3.2. Experimental setup

The experiment is conducted while immersing the bare surface of the plate in a bath of 3L of water, containing 100g of salt (NaCl). As shown in *Figure 14*, the wiremesh electrode is connected to the « CNTR » (counter electrode) terminal of the potentiostat through a multimeter set on ammeter mode. The mild steel plate is connected to the « WKG » (working electrode) terminal of the potentiostat and the SCE, after calibration, is connected to the « REF » (reference electrode) terminal. A « GND » (ground) terminal is also available to connect the material to the ground in order to protect it. The reference electrode must be fitted with a Luggin capillary and the end of the capillary must be placed as close as



Figure 14 : Potentiostat's terminals



Figure 15 : Experimental setup

possible to the bare metal surface. *Figure 15* shows the reference electrode, the Luggin capillary and the plate.

In *Figure 15*, the strike on the plate is under the water level and the Luggins capillary comes almost in contact with the bare metal surface. It is also important to notice the red wire connecting the covering anode to the « CNTR » terminal with an ammeter between the both elements.

4. Results

4.1. Covering anode : resistivity and consistency

A first set of samples were realized containing 15% wt., 20% wt., 25% wt. and 30% wt. of filler. The results were satisfying so a second set of samples were made containing 18% wt., 21% wt., 24% wt. and 27% wt. of filler. The resistivity was measured with a Fluke 45 Dual Display multimeter using several methods. The measure between two points was used first but it was considered as not suitable for this type of measure.

- Explanation on the measurement between two points

The measurement were done as shown in *Figure 16* and *Figure 17*. First, a measure was done between one fix point and a point distant from 1cm, then between the fix point and a point distant from 2cm until reaching the bottom of the plate. Each measured resistance was divided by the distance between the points. A second measure was done while measuring the resistance between two points distant of 1cm on several areas of the plate. The results and the calculation were reported in a table. The *Table 2*, shows a part of it.

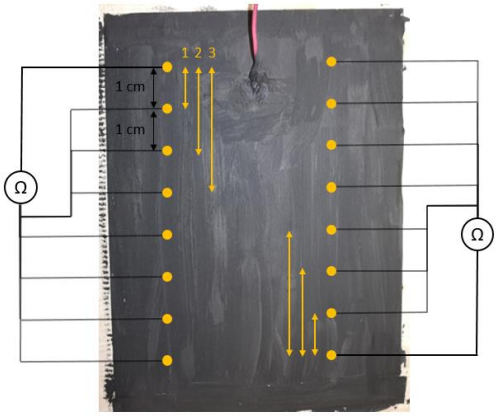


Figure 16 : Firsts measures between two points

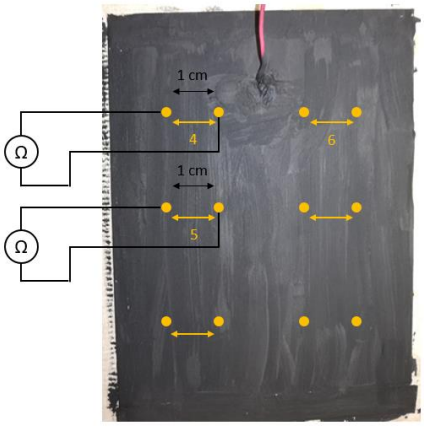


Figure 17 : Seconds measures between two points

Distance (cm)	1	2	3	4	Average (kΩ/cm)
Measured resistance (kΩ) Firsts measurements	2.3	2.8	2.6	4.2	/
Calculated resistivity (kΩ/cm)	2.3	1.4	0.9	1.1	**1.4**
Measured resistivity (kΩ/cm) Seconds measurements	3.6	2.0	2.2	1.6	**2.4**
Total average (kΩ/cm)		/		**1.9**	

Table 2 : Part of the resistivity full table

As shown by the results, the gaps between the measures and between the averages are too big. First of all, the unit Ω/m cannot be used to represent the resistivity neither for a volume nor for a sheet, it is used to define the resistivity of a wire for example. Indeed, the resistivity of a three dimensional material is defined by the following formula: $\rho = R \times \frac{w \times t}{L}$ in $\Omega \cdot \mathrm{m}$ (Eq. 9)

Where ρ : resistivity ($\Omega \cdot \mathrm{m}$) R : resistance (Ω) t : thickness (m)
w : width (m) L : length, current path (m)

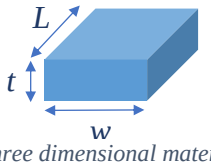


Figure 18 : Three dimensional material

Knowing this, it is easy to determine than a sheet resistivity is defined in Ω; the resistivity of a volume divided by the thickness. To avoid misinterpretation with the resistance, it is common to use the unit ohm per square: Ω/sq, which represent $\Omega \cdot \frac{\mathrm{m}}{\mathrm{m}}$.

The previous calculated resistivity could have been corrected while multiplying it by w, it would correspond to the sheet resistivity:

$\rho_{Sheet} = R \times \frac{t}{t \times L} \times w = R \times \frac{w}{L}$ in Ω/sq (Eq. 10)

Unfortunately, the width cannot be known in this case. *Figure 19*, shows the path of the current in this case of measurement.

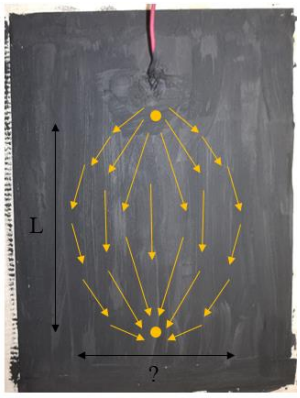


Figure 19 : Current lines while measuring the resistance between two points

Then, the results cannot be considered as representative of the reality. However, these measures were done on all the samples to verify the variation of the average values between the different samples. A qualitative interpretation of these values in *Table 3* can then be done.

Sample n°	Amount of filler (% wt.)	Resistivity (kΩ/cm)	Remarks on the consistency
1	15%	7370	Liquid before and during the application
2	18%	2809	/
3	20%	13.0	Thick liquid before and during the application
4	21%	9.0	/
5	24%	1.6	/
6	25%	1.5	Very thick liquid, dry during the application

7	27%	1.1	/
8	30%	2.0	Paste, dry quickly, apparition of granules during the application

Table 3 : Resistivity and consistence of the different covering anodes

The covering anode offering the best compromise between the conductivity seems to be *Sample 5*, with 24% of filler. Indeed, the conductivity is almost the same than *Samples 6*, *7* and *8*, and its consistency allows a good and uniform application. We can see that *Sample 8* has a higher resistivity than *Samples 5*, *6* and *7*, in spite of its higher amount of filler. This can be due to the irregularities on the covering as granules and variations of thickness.

These measures still gave two information:

- The coating is not uniform: some paths are very more conductive than others and some areas seems to be "isolated" from the rest of the coating.
- The resistivity decrease while the amount of powder increase.

The resistivity of the chosen covering anode was then measured with two other more accurate methods.

- ## Explanation on the two directs measurements methods

Figure 20 shows the measure of the resistivity directly done with two tapped copper wires and *Figure 21* shows the measure of the resistance done on a long line of anode.

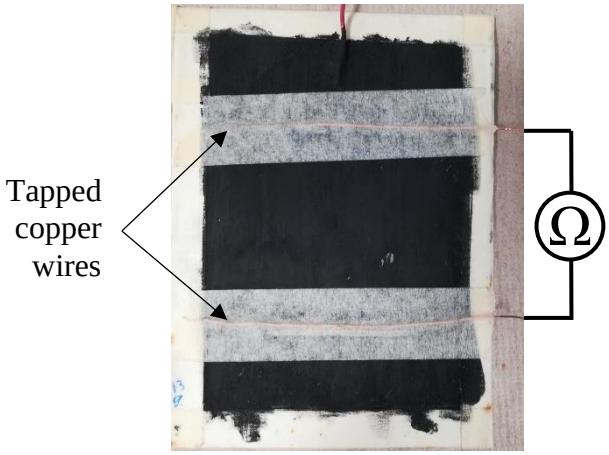


Figure 21 : Direct measurement of resistivity

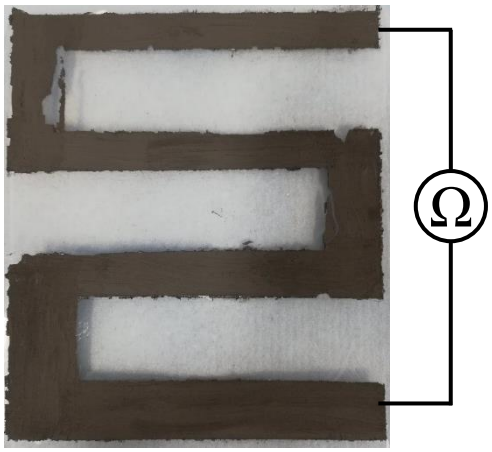


Figure 20 : Measurement of the resistance of the line

Using these methods, the calculation of the resistivity can be done because the width of the path taken by the currents lines is known; see in *Figures 22* and *Figure 23*.

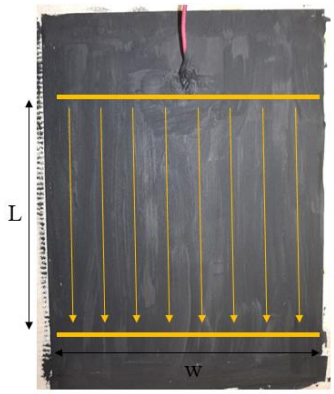


Figure 22 : Current lines while measuring the resistivity with the tapped copper wires

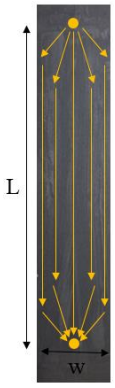


Figure 23 : Current lines while measuring the resistance through a long line of anode

Here, the calculation: $\rho_{Sheet} = R \times \frac{w}{L}$ can be done. *Table 4* gives the values of the resistivity of *Sample 5* obtained by the different methods. For the resistivity measured with the tapped copper wires there is no need to calculate anything because $L = w$. However, for the resistivity of the long line of anode, the value of $\frac{w}{L}$ is approximatively 0.0255 and the measured resistance is 101kΩ.

Resistivity measured using the tapped copper wires (kΩ/sq)	**Resistivity of the long line of anode (kΩ/sq)**
2.572	2.581

Table 4 : Resistivity of Sample 5

- ## Calculation of the resistivity through the circuit

The resistivity of the covering anode cannot be calculated using the electrical parameters through the full circuit: rectifier, anode, electrolyte and cathode. Indeed when the calculation of the resistance is done using Ohm's law, all the resistances in the circuit are included in the result: the resistance along the covering anode; R_A, the resistance between the anode and the electrolyte; $R_{A,E}$, and the resistance between the cathode and the electrolyte; $R_{C,E}$. The resistance along the electrolyte, R_E is negligible due to the very short distance between the two electrodes. As there is a need to know R_A to calculate the resistivity of the covering anode another method need to be used.

In the same way that the resistance of the covering anode was measured between the two tapped copper wires, a current flowing between those wires can be measured and knowing the potential applied, the resistance can be calculated. See the circuit in *Figure 24*. The recorded electrical parameters are shown in *Table 5*.

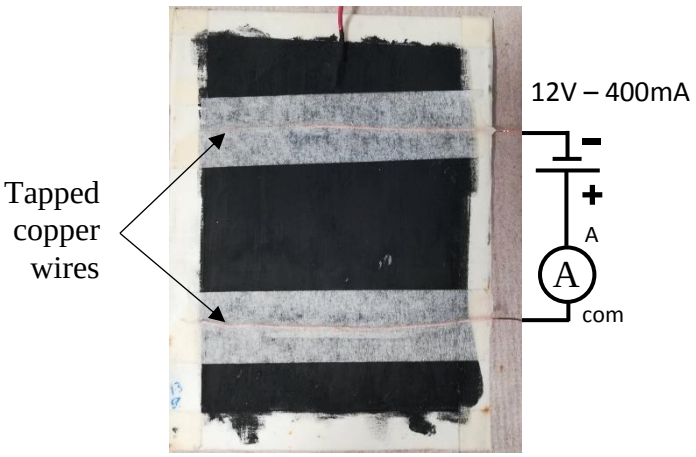


Figure 24 : Measure of the current flowing through the covering anode

4.2. Protection tests

A first experiment was done to determine the overprotection potential. The plate is said overprotected when hydrogen bubbles appear on the bare metal surface. The cathode was polarized at a potential of -1.5V versus a SCE. At this potential, a huge amount of bubbles appears on the metal surface; it is visible on *Figure 25*. At a potential of -1V, the bubbles remain but at a lower amount, almost invisible to the naked eye. However, no hydrogen bubbles were visible at a potential of -950mV. Other experiments were done while protecting the plate with potentials of -950mV, -850mV, -750mV and -650mV versus the SCE during 24 hours in salt water. *Table 5* presents the test parameters and the electrical properties of the cell, and *Figure 26* shows the different state of the metal surface after each test.

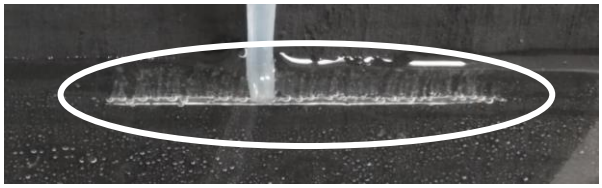


Figure 25 : Hydrogen bubbles on the bare metal surface

Measure	**Potential supplied by the rectifier**	**Net current**	**Time**
Instrument	Multimeter	Potentiostat / Multimeter	Potentiostat
Symbol	*U* (mV)	*I* (µA)	*t* (h)
Test 1	**Settings : E_{prot} = -950mV versus SCE – Time = 24h**		
	1976	94	2
	1905	83	10
	1917	63	24
Test 2	**Settings : E_{prot} = -850mV versus SCE – Time = 24h**		
	1650	35	2
	1717	48	10
	1762	57	24
Test 3	**Settings : E_{prot} = -750mV versus SCE – Time = 24h**		
	1418	52	2
	1489	50	10
	1510	49	24
Test 4	**Settings : E_{prot} = -650mV versus SCE – Time = 24h**		
	23	-300	2
	87	-271	10
	129	-217	24
Multimeter :	Mastercraft Digital Multimeter 52-0052-2		
Potentiostat :	Advanced Measurement Technology, Inc. Potentiostat/Galvanostat Model 273A		

Table 5 : Potentiostatic tests settings and data

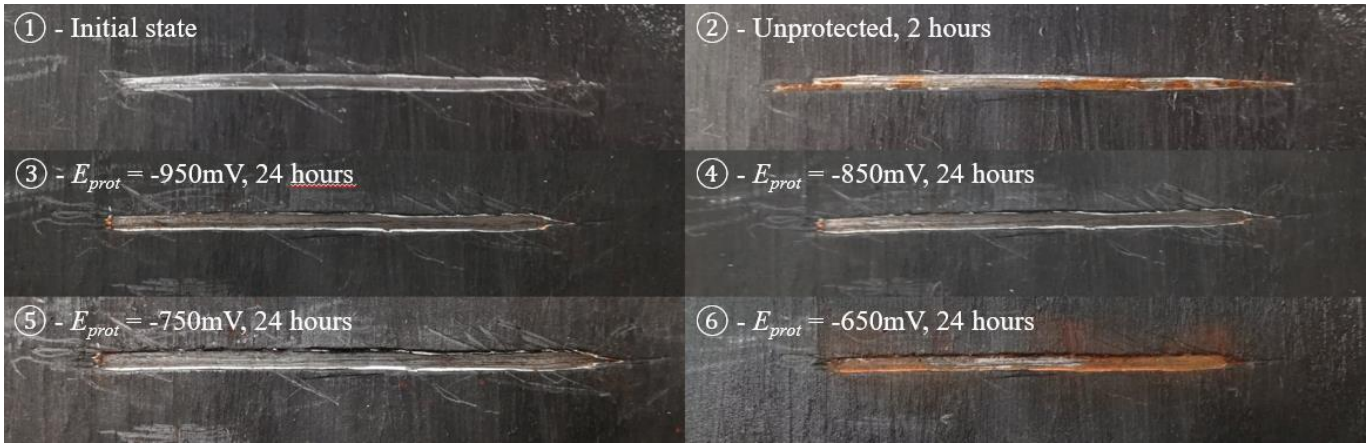


Figure 26 : Strike after different protection tests

The tests demonstrate the efficiency of the protection to reduce the corrosion rate. Indeed, picture ② of *Figure 26*, shows that within two hours of immersion without protection, the strike was already corroded. However, pictures ③, ④ and ⑤ show that while the protection potential is between -950mV and -750mV versus a SCE, the plate can spend 24 hours in immersion in salt water without getting significantly corroded. However, it is impossible to say that the plate is well protected with these potentials. For this, longer tests must be done to see if no corrosion nor blister occur after a week, a month and a year. Picture ⑥ shows that the plate is not well protected while -650mV versus the SCE is applied as protection potential. Moreover, the direction of the current changes (see *Table 5*), which means that the metal plate became anodic. Other studies demonstrated that a potential of -850mV against a SCE is a reliable as protection potential. *Table 6* shows the protection potentials against other reference electrodes.

SHE	Ag/AgCl	SCE
-608mV	-805mV	-850mV

Table 6 : Protection potential versus different reference electrodes

The potential to be applied by the rectifier to set the potential between the cathode and the SCE at -850mV to protect an area located 7cm away from the electrode is then between 1.65V and 1.76V. The consumed current after 24 hours to protect around 1cm² of bare mild steel is 57µA.

5. Conclusion

The covering anode technology solves many problems encountered by the common method and highly simplify the design of the ICCP systems:

- The covering anode behaves as a Faraday's cage to the stray currents: there is no risk of deteriorating the protected structure or surrounding installations.
- There is no need to dig anode beds nor to have huge anodes so the installation costs are highly reduced.

- The soil or water properties are not taken into account when designing the ICCP systems using the covering anode technology which reduces the study costs.
- The complicatedly shaped structures which are usually difficult to protect can be easily covered by a covering anode.
- The consumed current is highly reduced which reduces the working and installation costs.

This research also shows more specific information about the used covering anode and the protection parameters:

- A too high concentration of filler makes it impossible to apply a homogeneous layer.
- A too low concentration of filler leads to an extremely high resistivity.
- A protection potential more negative than -1V versus a SCE will lead to hydrogen damages.
- A protection potential less negative than -650mV versus a SCE does not provide a good level of protection.

Future experiments on this technology should examine the voltage drop along the covering anode. Indeed, this is the parameter which defines what area a wiremesh electrode can cover.

List of abbreviations

AISI : American Iron and Steel Institute
DC : Direct Current
ICCP : Impressed Current Cathodic Protection
SCE : Saturated Calomel Electrode
SHE : Standard Hydrogen Electrode

List of symbols

% wt. : Percentage in weight
°C : Degree Celsius
Ω : Ohm
A : Ampere
Cu : Copper
e^- : Electron
E_{corr} : Equilibrium potential
$E_{Fe/Fe^{2+}}$: Free potential of the couple Fe/Fe^{2+}
E_{H^+/H_2} : Free potential of the couple H^+/H_2
E_{prot} : Protection potential
Fe : Iron
g : Gram
h : Hours
H : Hydrogen
HCl : Hydrochloric acid
i_a : Anodic current density
i_c : Cathodic current density
i_{corr} : Corrosion current density
$i_{0\ Fe/Fe^{2+}}$: Current density of the couple H^+/H_2 at equilibrium
$i_{0\ H^+/H_2}$: Current density of the couple Fe/Fe^{2+} at equilibrium
L, l : Litre
$M^{n+}_{(aq)}$: Metal ion in aqueous form
$M_{(s)}$: Metal in solid form
m : Meter
NaOH : Sodium hydroxide
O : Oxygen
Ox : Oxidant specie
Red : Reductant specie
sq : Square
t : Time
V : Volt

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